

AISP programmable ionosonde

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Abstract

This paper presents a description of a prototype development project for a pulse compression ionosonde designed for vertical surveys using coded pulse radar. The project exploits Software Defined Radio (SDR) devices in an effort to provide a programmable instrument that is more versatile and advanced compared to traditional ionosondes. The programmable ionosonde (AISP) represents an evolution of the AIS (Advanced Ionospheric Sounder) ionosonde, which was entirely designed and built at INGV and is currently operational at several ionospheric observatories. The AISP ionosonde is intended for vertical soundings and primarily consists of an SDR device serving as both transmitter and receiver of Radio Frequency (RF) signals. Additionally, it generates all necessary control signals for the ionosonde's operation, including those for the RF power amplifier and antenna. The SDR device communicates with a Personal Computer (PC) via an Ethernet port, allowing for a wide range of configurations and enhancing the versatility of the ionosonde. Vertical surveys are conducted to map the ionosphere in specific regions, facilitating the validation of theoretical models of the global ionosphere.

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1. Introduction

The AISP Ionosonde (Zirizzotti et al., 2022) represents a significant advancement over the AIS Ionosonde (Arokiasamy et al., 2002), initially developed and constructed within our institute (INGV), and currently operational at ionospheric observatories worldwide, including Rome, Gibilmanna (IT), and others in countries like Spain, Argentina, Vietnam, and Kenya. However, over time the need emerged to bring this ionosonde up to date with recent advances and enhancements in radio frequency technology.

The AIS ionosonde uses complementary phase coded radar, emitting radio frequency pulses (generally in the HF frequency range from 1 to 30 MHz) and measuring the delay between transmitted and received pulses to determine the distance to the ionized layer reflectors in the ionosphere. This technique enables the measurement of reflective layer heights in the ionosphere at the sounding frequencies to obtain “ionograms” (Arokiasamy et al., 2002). Ionospheric soundings can be “vertical”, when the radar receiver and transmitter are in the same place (monostatic radar), or “oblique”, when the receiver and transmitter are at different sites (bistatic radar). Vertical soundings are affected by issues during transmission due to direct exposure of the receiver to the high power signal, which can cause saturation or damage to the receiver's initial stages, necessitating deactivation during transmission (“blind time”). In contrast, oblique soundings (bistatic radar) are not affected in this way because the transmitter

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and receiver are separated, enabling use of techniques like continuous wave (CW) radar (Farnett and Stevens, 1990; Alter and Coleman, 2008). However, bistatic radar over large distances requires precise temporal synchronization of transmitted and received signals, achieved through GPS systems, ensuring synchronization errors less than 50 ns smaller than a microsecond. GPSDO techniques allow for synchronization of transmitter-to-receiver reference oscillator phases with minimal errors.

Advances and miniaturization in electronic devices have enabled the development of components (mostly digital) capable of operating at higher frequencies. Faster analog to digital converters (ADCs) and digital to analog converters (DACs), operating at GHz frequencies with increased bit depths (up to 24 bits), alongside fast processors, have largely supplanted analog systems for radio frequency (RF) signals. The integration of programmable logic (PLA, CPLD, FPGA) has further simplified the design, construction, and testing of complex systems. In the 1980 s, Digital Signal Processing (DSP) introduced real analog signal processing capabilities, initially in the audio domain, converting analog signals to digital for numerical processing, including Fast Fourier Transform (FFT) techniques, before reconverting to analog signals. Over time, RF signal processing became feasible as conversion and processing speeds increased. This transition has addressed many challenges inherent in analog RF electronics, such as coupling, noise, and non-linearity.

However, a critical aspect remains the need for high speed communication ports (e.g., 1 Gb/s Ethernet) to transmit digitized RF signals to memories and processors. Such high speed communication is crucial since it defines a device's bandwidth, which is essential for accurately reproducing signals without introducing distortions.

Software Defined Radio (SDR) devices exemplify the innovation driven evolution of radio systems (Stewart et al., 2015; Rouphael, 2009). These devices facilitate the creation of intricate modulation and demodulation setups for radio signals with minimal programming effort. By drastically reducing the need for complex electronic circuits to process analog signals, SDR devices can be applied in diverse systems like telecommunications, RADAR, Bluetooth, WiFi, and others. Their programmability enables rapid adjustment to system characteristics such as operating frequencies and bandwidth, even allowing for remote updates and enhancements.

Today, the market offers a wide array of SDR devices with varying features. Commercial SDRs, e.g., Ettus Research USRP (Universal Software Radio Peripheral) devices (web reference: [Ettus research products](#)), prove particularly suitable for ionospheric radars, thanks to their capability of probing frequencies from 1 to 30 MHz, and their adaptable bandwidth options. Moreover, standard PCs equipped with fast Ethernet ports can function as digital signal processors (e.g., Ettus Research SDR), while more budget friendly options utilize fast USB ports. When integrated into radar systems, these programmable devices

offer remarkable versatility. All modulation and intermediate frequency (IF) oscillators are programmable, facilitating seamless adjustment of transmission and reception frequencies. Digital conversion techniques enable easy programming of signals and coding, simplifying transitions between different radar modes. Additionally, digital filters offer more flexibility compared to their analog counterparts, which are often complicated and tend to deteriorate over time.

Achieving comparable flexibility with analog techniques has proven to be arduous and costly, often necessitating intricate systems with extensive analog filter banks for different frequencies and codings. Analog filters are also prone to degradation, affecting operational efficiency and requiring frequent recalibration. The programmability of SDR makes it possible to configure an AISP ionosonde to transmit and receive various types of radar signals with different parameters.

The following paper provides an introductory section describing the theory of operation of the newly implemented radar system, followed by details of the hardware and software developed, and concludes with a description of the tests conducted to verify the correct operation of the ionosonde.

2. AISP: operating principles, characteristics and comparison

The new AISP ionosonde basically functions as a monostatic pulse radar, characterized by several key properties outlined below. The system emits RF pulses of finite duration, denoted as τ . When a pulse encounters an obstacle that reflects energy towards the receiving antenna location (typically the same site as the transmitting antenna, ignoring insignificant spatial differences), an echo is detected after a measurable time interval, t . This “flight time” can be used to determine the distance of the reflector, R , applying the simple equation: $R = (c \cdot t) / 2$, where c is the speed of light. It is important to note that the actual speed of electromagnetic wave propagation in the atmosphere, particularly in the ionosphere, can deviate from the speed of light, but for initial approximations it is commonly assumed to be equal to c . Subsequent determinations rely on more precise measurements conducted by the Ionosonde.

This type of radar exhibits distinct properties, derived from reliance on echo detection with a duration determined by the time for distance measurement. The spatial resolution, often called the “minimum distinguishable distance,” is denoted as R_{min} and directly influenced by the minimum pulse duration. Thus, the formula $R = c\tau / 2$ enables assessment of the minimum distinguishable distance between closely situated reflectors. Additionally, due to the inability to activate the receiver during impulse transmission to avoid saturation or damage, there is a minimum non-detectable distance, which is equal to the resolution. This phenomenon is termed the “blind range,” when the distance traveled by the pulse is denoted as $R_{min} = R$.

Echo detection depends on various factors, including received energy, which is influenced by the transmitted energy, signal attenuation, reflective properties of the target, and background noise (Blake, 1990; Zirizzotti et al., 2021). In the case of a simple impulse, the product of instantaneous power and duration determines its energy. Therefore, to avoid excessive power increases, extending the duration can be considered. However, an extension might reduce the system's resolving power. Theoretical studies indicate that resolving power is more closely tied to pulse bandwidth than duration and short pulses inherently have greater bandwidth compared to longer ones. Nevertheless, it is feasible to generate long pulses with broad bandwidths, comparable to short pulses, through suitable coding techniques, thus maintaining high resolution.

One of the simple coding systems, applicable even with modern digital techniques, involves dividing a transmitted pulse into N “sub-pulses” of duration T_s . Within each sub-pulse, the phase of the carrier is modulated based on a binary code, when each binary digit (0 or 1, or -1 and 1) corresponds to a specific phase, or the phase flipped by 180° . A simple example of coding is the so called 13 bit Barker (Farnett and Stevens, 1990) code (11111–1–1,1,1,–1,1,–1, 1), which generates the signal depicted in Fig. 1.

In the demodulation process, in theory (ignoring practical distortions) it is feasible to acquire a signal containing the sequence of the transmitted code. Subsequently, for recognition purposes, the received signal is processed for correlation (best matched filter) with the transmitted signal, resulting in a “compression” effect. This occurs because the lengthy transmitted pulse, lasting $N T_s$, is condensed into a pulse of the duration of T_s alone. However, without appropriate precautions, such as those applied with Barker codes, the “compressed” pulse may exhibit “lateral lobes” that could disrupt accurate concurrent recognition of multiple targets. Barker codes are designed to minimize side lobes with 13 bits as the maximum bit count.

It is possible to completely delete the sidelobes by transmitting more than one pulse in a Golay sequences (or complementary codes) (Golay, 1961; Farnett and Stevens, 1990). An example 16 bit complementary code is shown in Table 1.

The sequences are designed in such a way that during correlation (see Eqs. (7) and (8)), the lateral lobes have equal amplitude but opposite signs. Consequently, when the two echoes are added together, the lateral lobes cancel each other out. Fig. 2 illustrates the correlations of the two sequences and their sum.

The correlation process improves the signal to noise ratio of the received signal. In fact, the amplitude of the central peak of the correlation, denoted as A_{cor} , increases by a quantity that is dependent on several factors, including the amplitude of the received signal (A), the number of bits in the code (N_b), the sampling frequency of the signal (f_s), the duration in seconds of the sub-pulse (T_s), and external noise. In the ideal case of absence of external noise, A_{cor} can be written as:

$$A_{cor} = A * K_{cor} = A * N_b * T_s * f_s = A N_b N_s \quad (1)$$

Note that when using a complementary code, with the need to add two correlations at the end (see Eq. (8)), the result of Eq. (1) is multiplied by two and that the product $T_s * f_s$ represents the number N_s of points acquired within the sub-pulse.

The maximum amplitude of the side lobes (A_s) is calculated as $A_s = A * T_s * f_s = A * N_s$. This implies that the signal-to-noise power ratio (in the best case considering the side lobes as noise) for the Barker code mentioned earlier is $SNR = N_b / 2 = 169$ (equivalent to 22 dB). In Eq. (1), $K_{cor} = N_b * N_s$ is the gain of the correlation process, which manifests as a higher amplitude in the received signal. For instance, for the 16 bit complementary code in Fig. 2, sampled at a frequency of 200 kHz with sub-pulses of 30 μs (see Table 2), a gain $K_{cor} = 192$ (equal to a gain of 46 dB) is achieved.

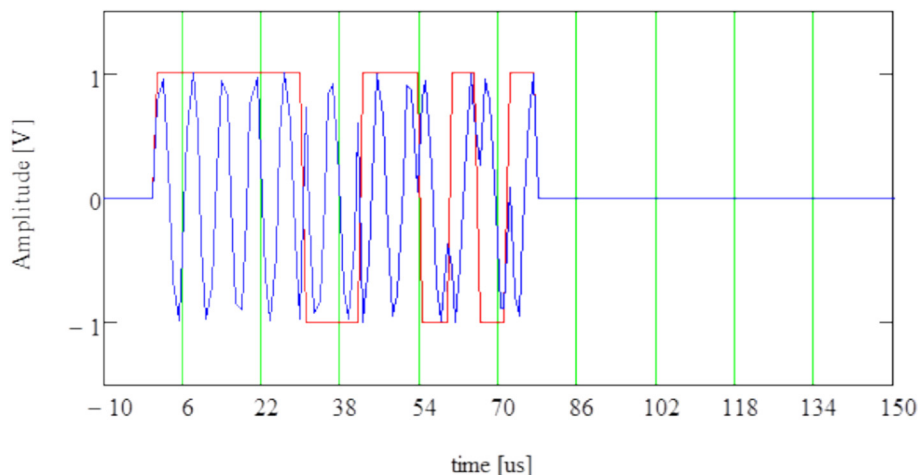


Fig. 1. Phase encoding of a 13 bit Barker code.

Table 1
16 bit complementary code.

Code order	Bit sequence
Code 1	1,1,1,-1,1,1,-1,1,1,1,-1,-1,1,-1,1,-1
Code 2	1,1,1,-1,1,1,-1,1,-1,-1,1,1,1,-1,1,1

The spatial resolution is ultimately determined by the length of the pulse after compression and is derived from the previously mentioned expression, except that the actual pulse duration should be substituted with the duration of the compressed pulse, which is the duration of the sub-pulse (T_s):

$$\Delta R = \frac{cT_s}{2} = \frac{c}{2B} \quad (2)$$

In the last step, the sub-pulse duration was substituted with the bandwidth B , which, at its limit, equals the reciprocal of the duration. As will be discussed later, when transmitting a signal with a sub-pulse duration of T_s , the system's bandwidth must be slightly wider than its reciprocal ($B > 1/T_s$) allowing the signal to transfer without distortions in order to achieve the minimum resolution.

The performance of pulse radar can be usefully compared with FM-CW (“chirp” radar) (Farnett and Stevens, 1990), an alternative radar system used for ionosondes (Bibl, 1998), by contrasting the two radar resolutions. In chirp radar a continuous sweeping radio frequency signal is transmitted, generally from 1 to 20 MHz. The chirp resolution is determined by Eq. (3)

where c represents the speed of light and α denotes the frequency sweep rate in Hz/s):

$$\Delta R = \frac{c\Delta f_b}{2\alpha} = \frac{c}{2\alpha T} \quad (3)$$

The parameter Δf_b corresponds to the range of frequencies used for scanning, while T represents acquisition time. Although the maximum interval can extend to cover the entire scan, approximately 15 or 20 MHz, and the corresponding time can reach values of around 30–40 s, practical observations of reflections at different heights, corresponding to various carrier frequencies, limit the effective utilization of the entire frequency range. Furthermore, assuming there was only one layer, the resulting resolution would be excessively high (around 10 m over distances from 90 to 1500 km). Therefore, during processing, the frequency scan is divided into a certain number of sub-intervals, denoted as NF (about 150–200), with each interval treated separately. The amplitude of each subinterval, $\Delta f_b/NF$ (approximately 80 kHz), is small enough to reasonably assume that the reflection occurs at only one distance. The corresponding time is very short (about 160 ms), and the resolution reaches acceptable values, around 1.8 km.

However, the most significant consideration at this point is the interpretation of the coefficient preceding the term $c/2$, which is the dimension of time, in pulse radar corresponding to the duration of the pulse (either real or after compression). In this case, inserting the values yields a duration of approximately 12 μs , which can be interpreted as the duration of the “compressed” pulse, since this is

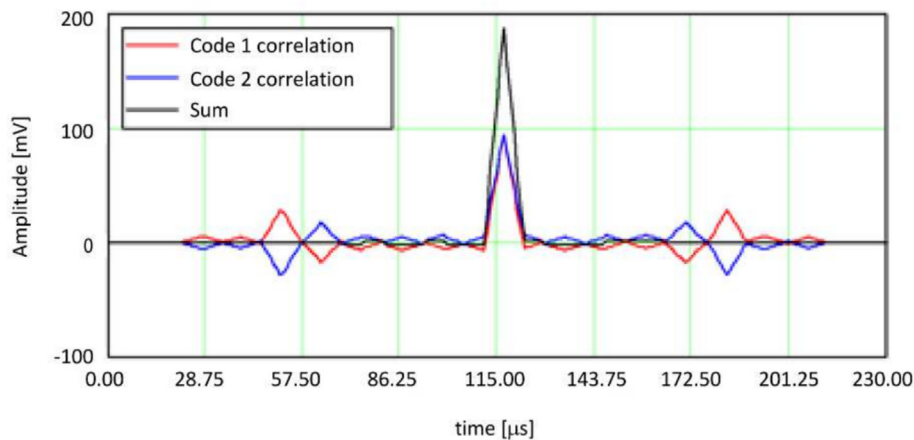


Fig. 2. Correlations for the 16 bit complementary codes and their sum.

Table 2
Comparison between different pulse compressions.

Radar type	No. bits	Subpulse duration [μs]	Pulse duration [μs]	Fs [kHz]	Compression factor	Resolution [km]	Process gain	Bandwidth [kHz]
Pulsed	1	—	480	200	1	72	1 (0 dB)	2.08
Barker code	13	37	480	200	13	5.6	96 (40 dB)	27.0
Complementary code	16	30	480	200	16	4.5	192 (46 dB)	33.3
FM-CW radar	—	12#	160,000	50	13,400	1.8	116 (41 dB)	81.8

effectively what determines the resolution, similar to the compressed pulse of a pulsed ionosonde. However, the actual duration was observed to be approximately 160 ms. Therefore the actual compression factor is approximately $\alpha T^2 \approx 13400$. The gain amplitude of the correlation of the chirp radar can be calculated (Hein, 2004) as $\sqrt{(\Delta f_b T) / NF}$. These observations explain the higher resolution of this type of signal processing system.

Table 2 summarizes the process gains and different resolutions achievable with various radar signal compression techniques (Farnett and Stevens, 1990; Zirizzotti et al., 2021), highlighting performance differences based on the same pulse duration. The table was compiled for comparative purposes and certain parameters (marked with #) need to be interpreted appropriately. For example, in the case of the complementary code, there are two pulses, so the actual duration to be considered would be double, but in this case there are no sidelobes (which could be considered as process noise) and the noise is the electronic noise of the receiver. Similarly, in the case of CW radar, the pulse duration is not real because there is no actual pulse, but only sub-intervals.

Regarding bandwidth, additional considerations must be made about the practical implementation of the system. As discussed above, the bandwidth occupied by the signal determines distance resolution. Constructing an analog broadband radar system is challenging because all the system components must operate within a broadband spectrum. However, the use of modern SDR systems greatly simplifies the task, with devices available today that facilitate the acquisition of the desired bands for ionosonde applications. It is important to note that, in addition to the device itself, the speed of the data communication port must also be taken into account.

When using a signal with a bandwidth of 200 kHz, it is necessary to sample at a frequency at least twice as high. If a D/A device is used to generate a 200 kHz bandwidth signal with sampling to ensure at least five samples for each sub-pulse, a sampling frequency (f_s) of 1 GHz can be achieved. Assuming integer data quantized at 8 bits, the

required transfer speed is 8 Gbit/s (1 Gbyte/s), which is now normal for Ethernet and USB ports.

It is also crucial to consider the bandwidth occupied by the signal during reception. The bandwidth occupied by a signal formed from a sequence of ideal pulses obtained by sampling a received signal is theoretically infinite. However, in practice the sampling frequency can be significantly higher than $1/T_s$ (when T_s is the duration of the sub-pulse) and the non-ideal nature of the sampling process results in a not infinite bandwidth. Instead, the energy tends to concentrate in a relatively modest range around zero frequency. The following empirical formula gives an evaluation of the bandwidth occupied by a coded pulse signal (regardless of the code used):

$$B_4 = \frac{4}{T_s} \quad (4)$$

Fig. 3 illustrates the normalized power spectrum for a 16 bit code ($T_s = 30 \mu s$). Calculating with the previous formula gives $B_4 = 133 \text{ kHz}$, indicating that approximately 98.1 % of the signal power falls within this bandwidth. The use of a low pass filter can effectively confine the bandwidth to this value without risking signal distortion or loss of detection capability. It is then also feasible to implement a higher sampling frequency to avoid aliasing issues.

Another crucial parameter to consider in ionosonde design is the Pulse Repetition Frequency (PRF), which is the inverse of the Pulse Repetition Time (PRT) and represents the time interval between two consecutive transmissions. This interval directly correlates with the maximum distance intended for investigation. Returning to the initial relationship $R = ct/2$, and setting $R = R_{\max}$, it can be determined that for $R_{\max} = 750 \text{ km}$, the corresponding $t_{\max} = 2 R_{\max}/c = 5.0 \text{ ms}$. This duration represents the minimum time required for accurately referencing the real distance of an echo originating from the maximum distance. Therefore, in this example the maximum PRF would be $\text{PRF}_{\max} = 1/t_{\max} = 200 \text{ Hz}$. The hardware of the AISP Ionosonde enables up to 100 Hz without issues,

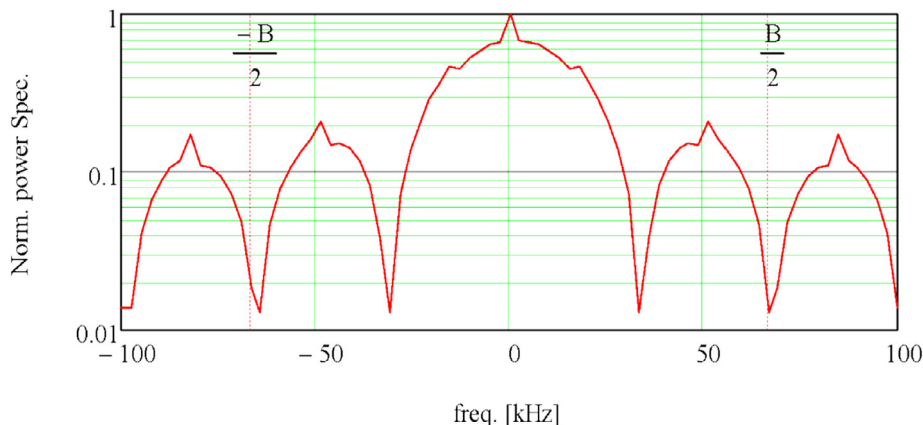


Fig. 3. Normalized power spectrum of a pulse signal encoded with $T = 30 \mu s$.

which is compatible with the limits mentioned above, and so a PRT of 10 ms must be configured.

Another vital parameter to consider is the number of measurements acquired and subsequently averaged, which serves to enhance the signal-to-noise ratio. A practical approach to this averaging procedure involves combining multiple echoes after correlation (at a defined frequency) but before module extraction (coherent integration). It can be demonstrated that the signal-to-noise ratio, defined as the ratio of signal power to noise power, improves by a factor of M with this type of integration, when M is the number of traces acquired for a given frequency.

Once M is determined, the total time required for a survey can be calculated. For the AISP Ionosonde, the total range of frequencies to be probed was selected from $F_{ini} = 1$ MHz to $F_{fin} = 20$ MHz, with a frequency resolution of $F_{res} = 0.05$ MHz. Consequently, a total of 381 traces (frequencies) are acquired. By setting $M = N_{AVG} = 30$ as the integration factor, the probing time can be calculated as follows:

$$T_{sound} = PRT \frac{F_{fin} - F_{ini}}{F_{res}} N_{AVG} = 114s = 1'54'' \quad (5)$$

If a double (complementary) code is employed, the probing time doubles to 3 min and 49 s. However, it is essential to note that the AISP system has a specific requirement: to start acquisition 0.1 ms before the transmission pulse (equivalent to 10 samples), even though this adjustment minimally reduces the maximum achievable distance.

The parameters outlined thus far, along with numerous others influencing the system's operation, can easily be modified by reprogramming the ionosonde's SDR device. [Table A1](#) included in the Appendix summarizes the characteristics of the AISP ionosonde.

3. Coherent correlation

As already mentioned, coherent integration allows the signal to noise ratio to be improved. [Fig. 4](#) is a block diagram for received signal processing. The leftmost section depicts the “in phase” and “quadrature” acquisition technique for a signal originating from the antenna, as executed

by the SDR device. During the sampling process coherent demodulation of the so-called I/Q components (in-phase and quadrature) is conducted simultaneously. This demodulation is achieved with a carrier frequency, f_c , which is bifurcated into two branches differing in phase by 90° .

This acquisition method offers several advantages ([Schreier and Scharf, 2010](#)). For example, it eliminates the need for a subsequent “notch” filter to remove spurious frequencies during complex signal demodulation ([Proakis and Manolakis, 2021; Zirizzotti et al., 2021](#)). Phasor multiplications do not include frequency sum and differences, unlike Werner trigonometric formulas. Moreover, I and Q acquisition enables complex correlation of signals while retaining both module and phase information. Furthermore, opting to compute correlation in the time domain was strategic for reducing calculation time. This method provides results equivalent to frequency domain correlations but imposes greater processing demands for computing the Fourier and inverse Fourier transforms. Operating in the frequency domain could be more advantageous, but only when there is a need to filter the signal at specific frequencies. The complex correlation $Ccor$ in the time domain is defined as:

$$Ccor_n = \sum_{k=0}^{code_{length}} sign_{n-k} code_k \quad (6)$$

where $sign_n = sign_{In} + j sign_{Qn}$ is the complex received signal, $code_k = code_{1k} + code_{2k}$ is the transmitted code in complex form, and the asterisk represents its complex conjugate. Transmitting first code one and then code two successively into the receiver generates separate traces for codes one and two, making it convenient to correlate them separately:

$$\begin{aligned} Ccor1_n &= \sum_{k=0}^{code_{length}} sign1_{n-k} code1_k \\ Ccor2_n &= \sum_{k=0}^{code_{length}} sign2_{n-k} code2_k \end{aligned} \quad (7)$$

Note that signals 1 and 2 are complex while the codes are real signals. The sum of the correlations of the two complementary codes allows the lateral lobes to be deleted as shown in [Fig. 2](#):

$$Ccor = Ccor1_n + Ccor2_n \quad (8)$$

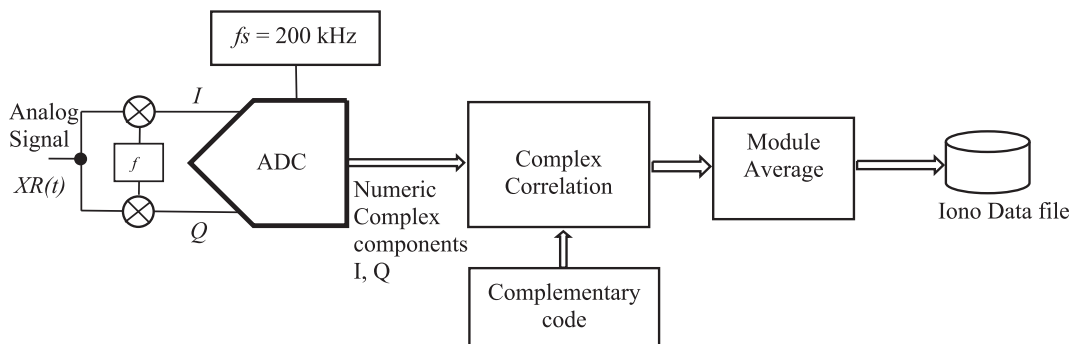


Fig. 4. Schematic of the digital processing of a received signal.

The average of the resulting complex correlations enables coherent integration of the received signals. This process averages the amplitudes and phases of the signals, leading to a significant improvement in the resulting signal quality. After averaging, the signal modulus is calculated so that the acquired data can be plotted in ionograms.

4. AISP software

The AISP Ionosonde software comprises two main elements: the Ionosonde operating software, and the acquired data analysis and conversion software. The operating software configures and communicates with the SDR device, as well as processing the data transmitted and received during communication. It is coded mainly in C++ and relies on the [Ettus Research UHD](#) (USRP Hardware Drive) library (web reference: [Ettus research UHD Manual](#)) for its device configuration and communication functions. It is currently compatible with machines running the Ubuntu 21.04 operating system. However, it could also be configured to run on Windows and Mac OS machines ([Zirizzotti et al., 2021](#)). The numerical techniques implemented in the AISP Ionosonde, leveraging SDR devices, involve capturing the entire time scan for a duration represented by $t_{\max} = N_{\text{sample}}/f_s$. This process entails sampling both data series, denoted as I and Q, from the received signal.

[Fig. 5](#) provides a detailed flow diagram for the AISP Ionosonde program. The program operates in command-line mode and accepts a large number of radio sounding configuration parameters as input. This flexibility empowers users to explore diverse radio sounding techniques with the use of different waveforms including pulses, chirps, phase codes, and complementary codes. The program automatically loads default parameters for standard operation, minimizing the need for extensive digitization. After reading and validating the input parameters, the program configures the digital lines of the SDR, facilitating the generation of digital control signals for the RF power amplifier (Amp. Trigger) and receiver trigger. This ensures that the receiver closes during transmission and opens during the radar echo reception phase. The configuration of the SDR includes sampling frequencies for both reception and transmission, as well as defining central frequencies for modulation and demodulation.

The program also facilitates configuration of various parameters for SDR models equipped with different “daughter boards.” These boards may incorporate additional filters or numerous transmission and reception channels. The transmitted pulse is constructed digitally by populating the “wave_table”, an array consisting of 8192 complex numbers representing the signal’s numerical values for the waveform being used. Users can select different waveforms from the command line, ranging from simple pulses, chirps, phase codes, and complementary codes, specifying parameters such as duration, pulse repetition rate (PRR), and signal amplitude.

In the case of complementary codes, the “wave_table” is expanded and the two codes are written consecutively as real numbers. The data is subsequently filtered through an anti-aliasing low pass Finite Impulse Response (FIR) filter, before being transmitted to a Digital-to-Analog (DAC) converter where it is converted into an analog signal at the transmitter’s data sampling rate.

The filtered values are simultaneously used for correlating the two codes. The computer allocates memory buffers to store received trace data, correlation results, and averaged values. A data file is then created in *.RDF binary format, named according to the date and time, with radio sounding parameters detailed in its header and following a format similar to that of the AIS Ionosonde. The transmission and reception cycle commences, spanning the range from the initial to final designated radio sounding frequencies. Traces at various frequencies with distinct codes are iterated for the specified number of averages and stored in the receive buffer for subsequent processing. Upon completion of radio sounding, complex correlations of the received signal are computed. Correlations of the complementary codes are summed and subsequently averaged. This processing approach ensures maximum coherence between the averaged signals, provided that an excessive number of averages are not utilized, which could result in measurements extending beyond the coherence time of the ionosphere (typically around ten seconds). The program concludes by calculating the magnitudes of the complex values and saving all the processed data.

Precise and repeatable generation of a transmitted pulse and signal reception are available in the UHD library thanks to the “timed command,” which can be issued to most Ettus USRP devices. This can be used to specify precise times for data stream commands, configure gains, set centre frequencies, switch antenna ports, and toggle GPIO lines (web reference: [Ettus research UHD Manual](#)).

5. Ionogram analysis and visualization software

Various Python scripts have been developed for ionogram data analysis and visualization. The `Ionogram_02` package uses two sets of Python scripts to filter the acquired data. One extracts data from raw signals and the other compresses the data for the “Autoscala” process ([Scotto and Pezzopane, 2002](#); [Pezzopane and Scotto, 2007](#)). The RDF files are stored according to creation date, organized in folders by year, month, and day. The script uses this format to navigate to the appropriate folder and, by default, processes the latest file. The acquired data ranges from 1 to 18 MHz with a step frequency of 50 kHz, resulting in 1000 data points sampled at a rate of 200 kHz. This data is compressed for virtual heights between 0 and 750 km with a resolution of 0.75 km. However, due to coupling, data points up to 90 km are removed by the software.

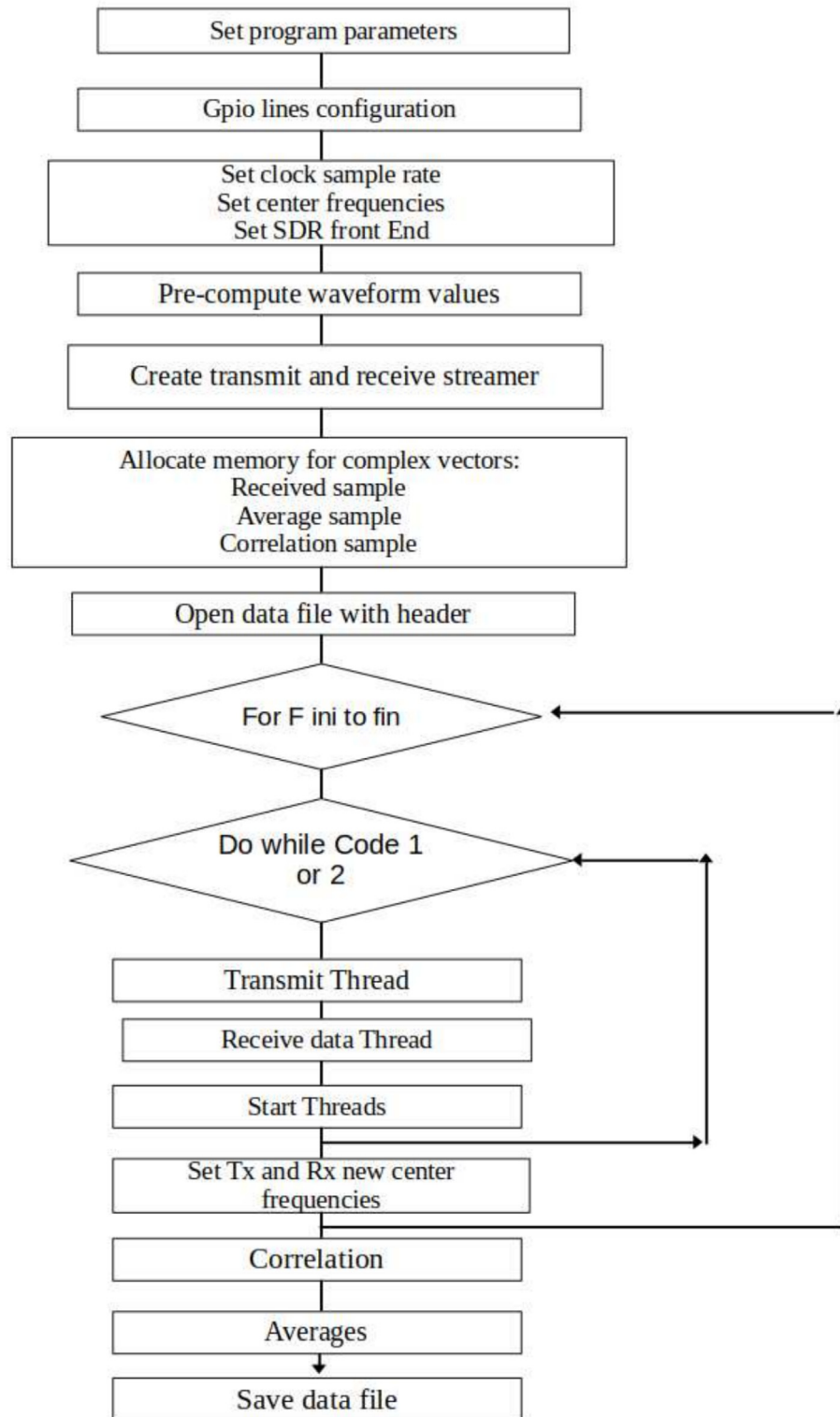


Fig. 5. Program flow chart.

Each filter cycle reduces a component of noise and extracts useful information. A single frequency component or single trace from AISP data is processed and each filtering step is plotted in Fig. 6. In Fig. 6a certain points in the

signal trace are marked with black circles, representing a target signal which should be filtered and plotted.

This is achieved using the 2D-CA-CFAR algorithm (Molina et al., 2016). The method was chosen because

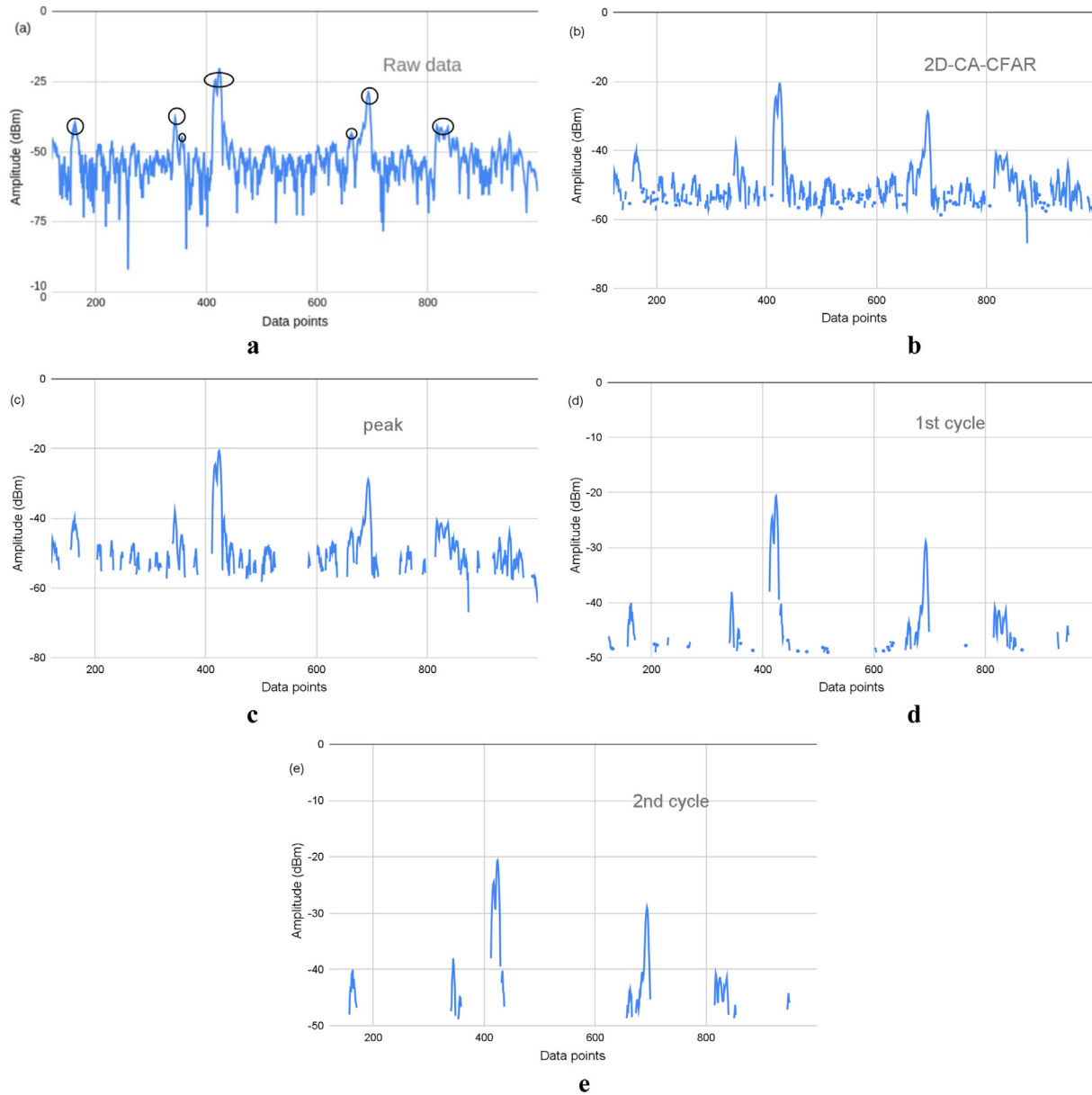


Fig. 6. Single trace data set after various filtering processes. (a) Raw data obtained by SDR, while (b, c, d, e) are data received after different filtering steps.

CFAR can shortlist valid targets, and by filtering two dimensionally only three dimensional absolute peaks are able to pass, with the result shown in Fig. 6b.

A 9×9 Matrix, shown in Fig. 7, is used as a sliding window, containing width and height of unit 2 for both guard and training cells for a single CUT (Cell Under Test). The window slides vertically eliminating the CUT as a false target or noise if a threshold is exceeded, otherwise it is retained as undisturbed. The threshold is distinct for each individual CUT based on the average of its guard cell values. The modified CUT standard is considered for the next set of processes.

The data from this step is passed to the peak detecting algorithm. Comparing the data in Fig. 6b and e shows that most of the jitter signals have been removed. The algorithm

compares the amplitude and width of the maximum signal values to filter potential signals.

Numerous radio broadcasting stations generate a lot of vertical noise in ionograms. This is removed by implementing a vertical averaging filter that compares the average power from a trace against the power of its individual data points. A second order average filter is used to cycle through the system twice. Output from the first cycle is shown in Fig. 6d, and output from the second cycle is shown in Fig. 6e, which contains all the signal traces marked in Fig. 6a.

The main advantage of this filter mechanism is to extend system flexibility. The filters are designed to operate without applying standard threshold values, instead calculating values from the RAW data provided by the SDR. The

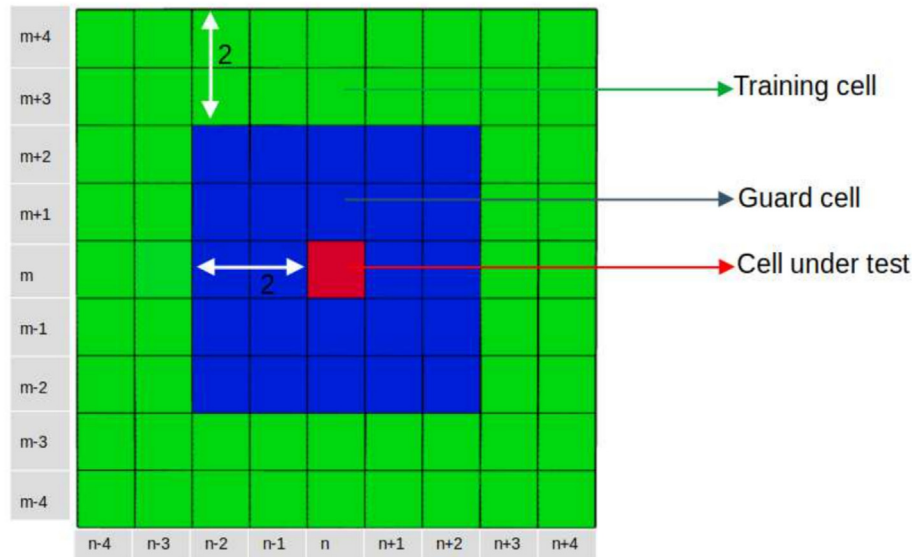


Fig. 7. Sliding window composition for 2D-CA-CFAR algorithm.

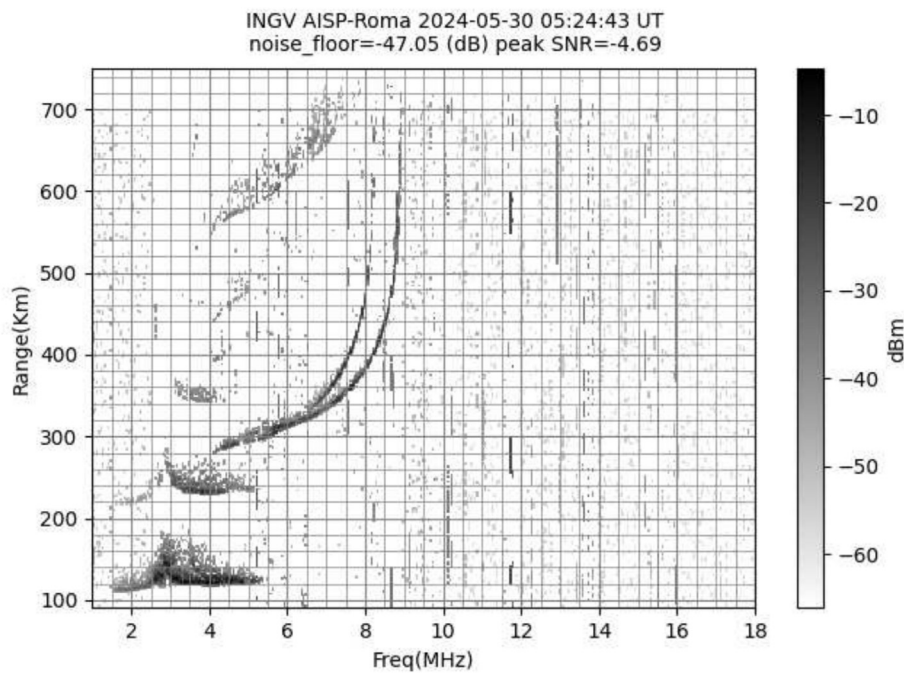


Fig. 8. Ionogram visualization.

result is the generation of usable ionograms, regardless of environment or location.

Fig. 8 shows an ionogram generated by the software. Grayscale coloring is chosen for enhanced visual clarity. The heading includes the plot creation date and time in universal time format, followed by noise floor and peak SNR.

Autoscala is a software developed at the INGV (Scotto and Pezzopane, 2002) to automatically scale ionospheric characteristics from ionograms. Autoscala required expansion to include a routine for the real time computation of the electron density profile, which is essential for ionospheric monitoring and space weather applications

(Scotto, 2009). Autoscala was initially used with ionograms recorded by AIS-INGV ionosondes located in Rome, Gibilmanna (Italy), and Tucumán (Argentina). These ionosondes lack the capability to record the polarization of received echoes. From its inception, Autoscala was designed to be adaptable to various ionosondes. This flexibility is due to its image recognition-based approach, which makes it independent of the hardware features of a specific ionosonde and capable of scaling ionograms without polarization tagging (Krasheninnikov et al., 2010; Pezzopane et al., 2010). Autoscala processes ionograms stored as binary RDF (Rough Data File) file format. Each

file consists of a 197 byte header followed by a set of records. The header contains critical information, including the initial and final sounding frequencies, frequency step, receiver settings, and site-specific geophysical parameters, such as geographical and magnetic coordinates, gyrofrequency, and polarization information. Each 150 byte record represents the energy reflected from heights ranging from 90 km (byte 1) to 760.5 km (byte 150), with a resolution of 4.5 km per byte. While AIS-INGV ionosondes natively produce files in this format, other ionosondes, such as AISP, require format conversion for compatibility with Autoscala. To perform such a conversion into the RDF format, it is necessary to compress the information by reducing data points to 150 and adjusting the resolution to 4.5 km. It is indeed necessary to ensure compatibility

with the standard 197 byte header and 150 data points per frequency.

As a first step, Autoscala uses routines to identify the main ionospheric characteristics such as foF2 (critical frequency of the F2 layer), foF1 (critical frequency of the F1 layer), and foEs (critical frequency of the Es sporadic layer). As a second step, in order to obtain the profile that best fits the recorded ionogram, an iterative technique is used to vary the parameters of a model within a range centered on some “base values.” These base values are calculated based on the ionospheric characteristics automatically obtained from the ionograms by the previously mentioned routines, taking into account the heliogeophysical conditions. The iterative procedure is carried out using the virtual heights measured at specific frequen-

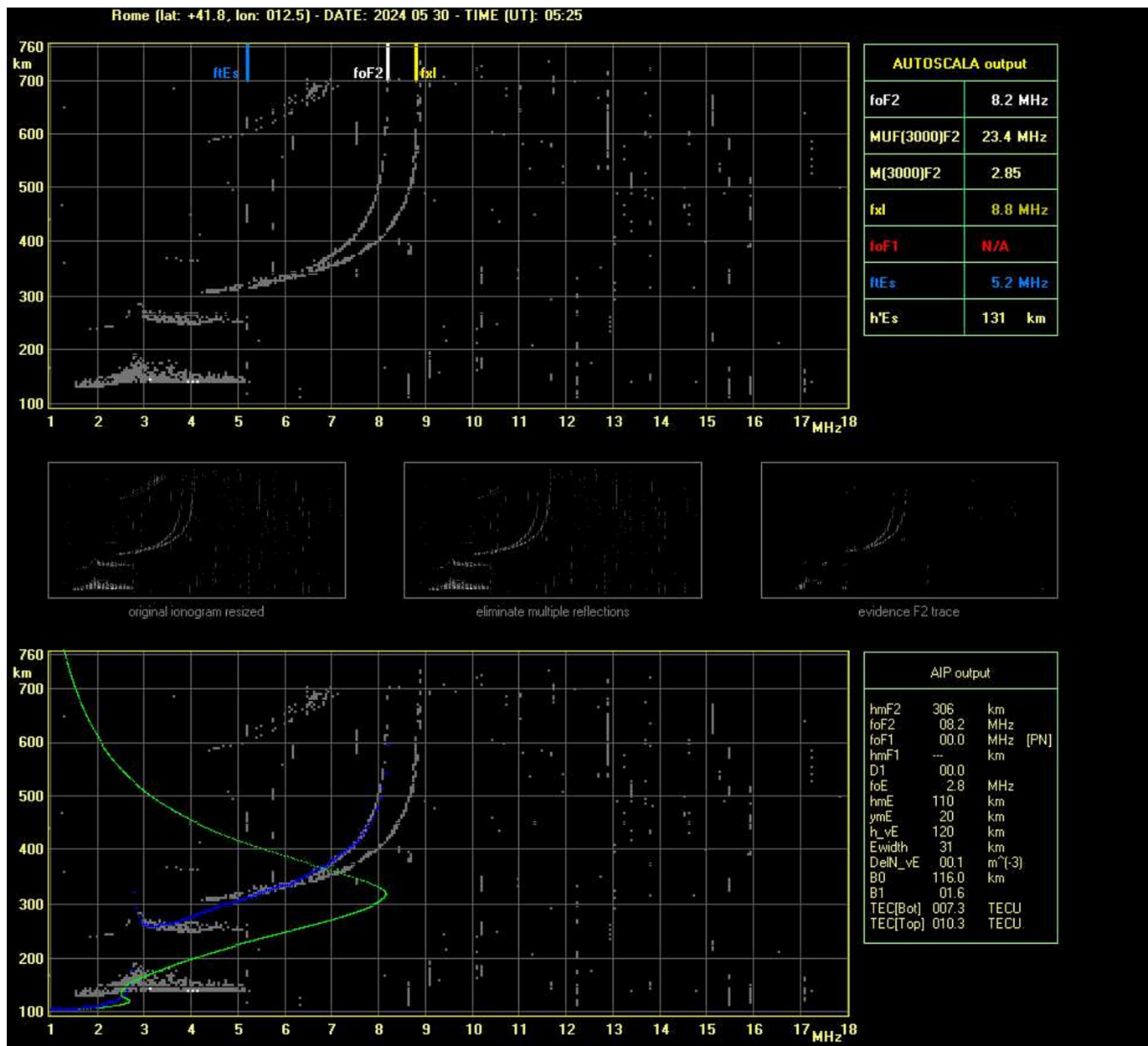


Fig. 9. Visualization of ionogram produced by Autoscala.

cies, while the reconstructed trace is displayed in blue across the entire frequency range (Scotto, 2009). In the example shown in Fig. 9, for frequencies below 4.2 MHz, the trace is reconstructed entirely based on the model, and the double reflection of the Es layer is not taken into account.

6. Hardware AISP

Fig. 10 is a block diagram for the AISP Ionosonde, indicating the connections between the hardware components (more technical information in Zirizzotti et al. (2022)). The design is relatively straightforward compared to traditional ionosondes because most of the signal processing is performed digitally by the computer.

The term “front end” refers to the interface designed for adapting external devices (antennas and power amplifiers) to the internal receiver or transmitter components. In the example, this involves the circuits of the SDR device. A limiter is installed on the receiver antenna to prevent signals of excessive amplitude from saturating the receiver input. A digital trigger signal (5 V) is generated by a GPIO (General Purpose Input/Output) in the SDR. The signal enables the receiver and transmitter during their respective phases, ensuring they are not activated simultaneously. The receiver trigger (GPIO line 2) is connected to a switch that completely disables the receiver during transmission. This prevents the transmitted signal being picked up by the receiving antenna and directly reaching the subsequent circuits, which could cause serious damage. The RF power amplifier trigger (GPIO line 3) activates the amplifier to transmit the signal to the antenna. An external RF power amplifier is included to boost the signal power sufficiently to reach the highest layers of the ionosphere.

The main system element is the Software Defined Radio (SDR). While the features and specifications of SDRs can vary between models, all these devices generally utilize the same architecture, as outlined in the technical report with relevant references (Zirizzotti et al., 2021).

The Ettus USRP N210 (web reference: Ettus research products) SDR device was used in the ionosonde design, offering a usable bandwidth of 25 MHz derived from its

daughter board, digital converters (ADC and DAC), and computer communication interface. Considering that each sample comprises 2×16 bits, plus various control characters, the 1 Gbit/s port speed determines a bandwidth of 25 MS/s for the device. The LFRX and LFTX daughter boards were selected because they are particularly suited to the frequencies and bandwidth required for operation of the AISP Ionosonde (for more information also see the web reference of Ettus research products; Zirizzotti et al. (2021)).

A fundamental hardware element for operating an AISP Ionosonde is the computer used to configure, receive, and transmit the operating data of the SDR device. This project used a PC with an 11th gen Intel i7-1165G7 @ 2.80 GHz $\times$ 8 microprocessor, 4 Gb of DDR3L RAM, running the Ubuntu 22.04.4 64 bit operating system, with a 256.1 GB SSD hard drive. The processing speed of the computer influences the operation of the SDR device, in particular the PRR, because the data exchange between PC and SDR must end before the next pulse is generated in order to avoid generating a “Late commands or packets” error for the timed commands.

7. Ionosonde test

The receiver front end shown in Fig. 10, was tested using an HP 8714 network analyzer. Receiver gain was measured during both the ON (receive) phase and the OFF (transmit) phase of the switcher (Analog Devices ADG901). Fig. 11 shows the two gain graphs, and Fig. 12 shows the input and output impedances of the receiver front end, which remain close to the standard 50 Ω in the frequency band used. Fig. 12 indicates a maximum gain of 11 dB during the reception phase, and an attenuation of less than -50 dB during transmission.

Two other types of tests were carried out on the AISP ionosonde. The first was a so-called “closed circuit” laboratory test in which a signal processed by a waveform generator is carried into the receiver, delaying transmission for a variable time in order to simulate echoes at different times and of different amplitudes. It was carried out using a simple radio frequency pulse because this is easier to obtain

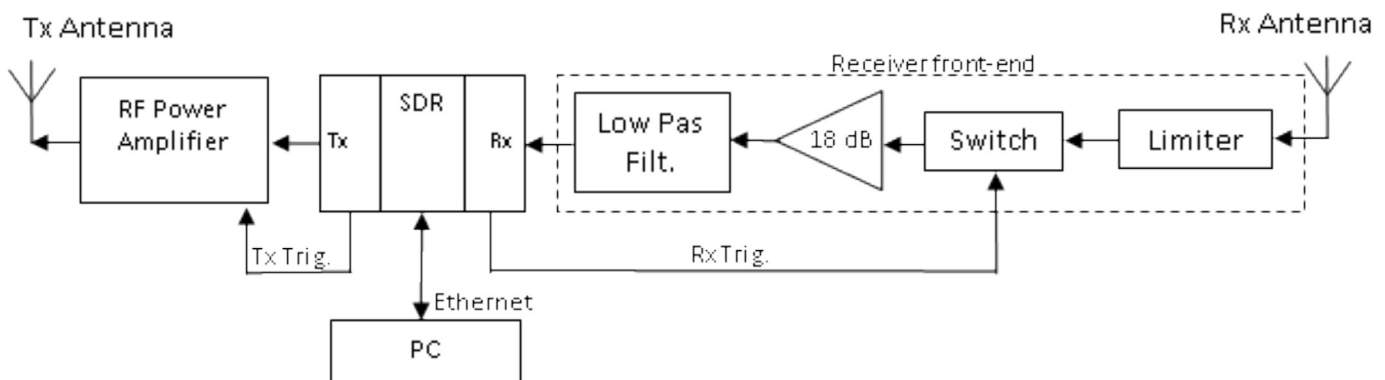


Fig. 10. Block diagram of the AISP Ionosonde.

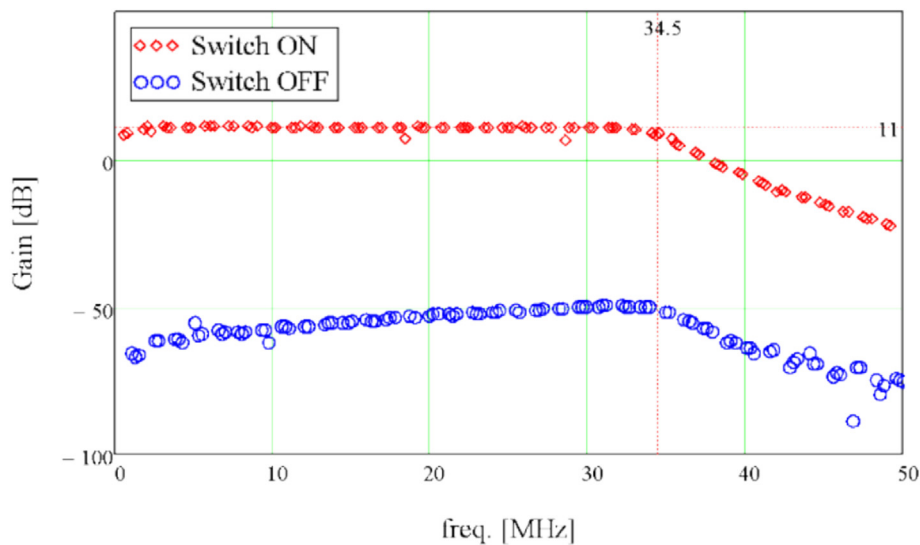


Fig. 11. Receiver transfer function.

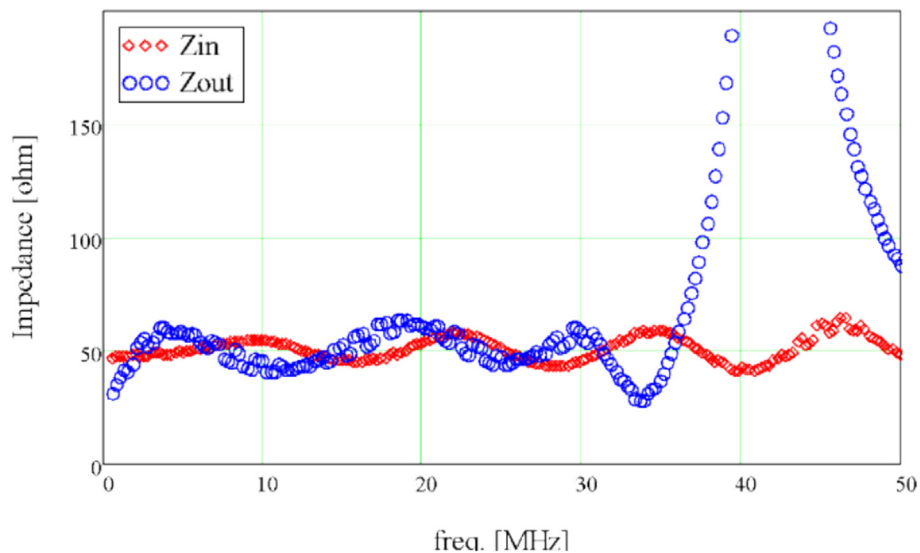


Fig. 12. Receiver input and output impedance.

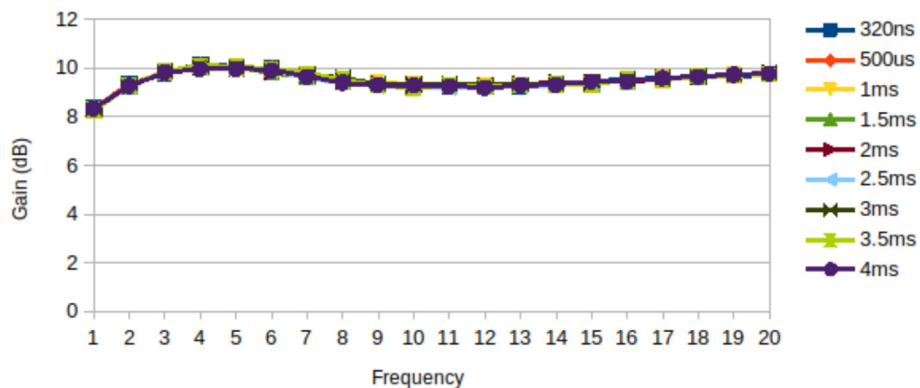


Fig. 13. Receiver gain vs. frequency for different delays.

from the laboratory's RF generators. This test allowed us to verify the overall operation of various parts of the AISP ionosonde and its internal receiver.

Some results from the laboratory tests, including the SDR input dynamics, are shown below. Testing also enables fine system calibration to ensure correct input signal strength. Fig. 13 shows frequency vs. gain measurements across the trigger circuit, at varying pulse delays defined using the Siglent SDG 830 generator, with the output measured using a standard oscilloscope. Clearly, the gain does not vary significantly with delay variation but instead varies with frequency. At 4 MHz, the amplification reaches its maximum of 10.13 dB with an average of 9.5 dB across the whole frequency band.

The minimum signal detectable by the receiver was about -90 dBm, enough to identify ionospheric reflections (Gilli et al., 2018). Calibration of the system is essential to provide a valid input scale and allow study of the absorption properties of the transmission medium. Fig. 14 shows the calibrated voltage to input voltage peak-to-peak. The SDR gain was set to 0, 3, and 6.5 dB using software. In Fig. 14, when 0 dB gain was implemented (yellow line), linearity was maintained throughout input range. However, under the other gain settings, the linearity was interrupted starting from about the 0.75 Vpp (1.5 dBm) level for the maximum gain of 6.5 dB. The maximum input power to the USRP N210 was 10 dBm (the maximum input signal for the N210 is 15 dBm).

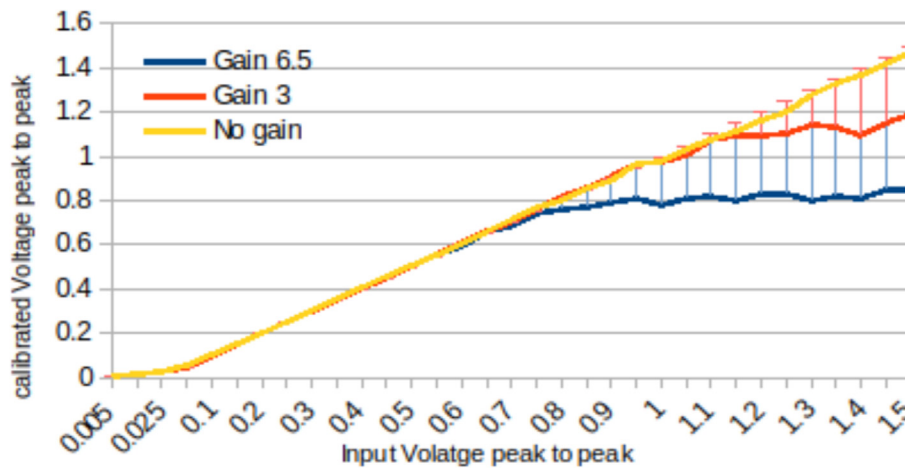


Fig. 14. Calibrated input for different rx gain settings.

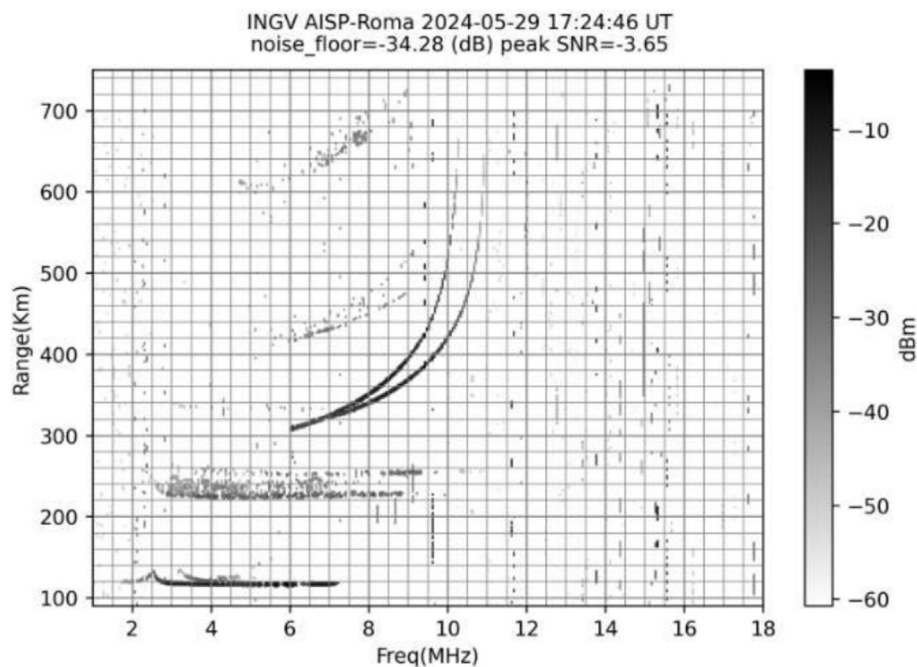


Fig. 15. AISP ionogram at 17:24 pm on 5/29/2024.

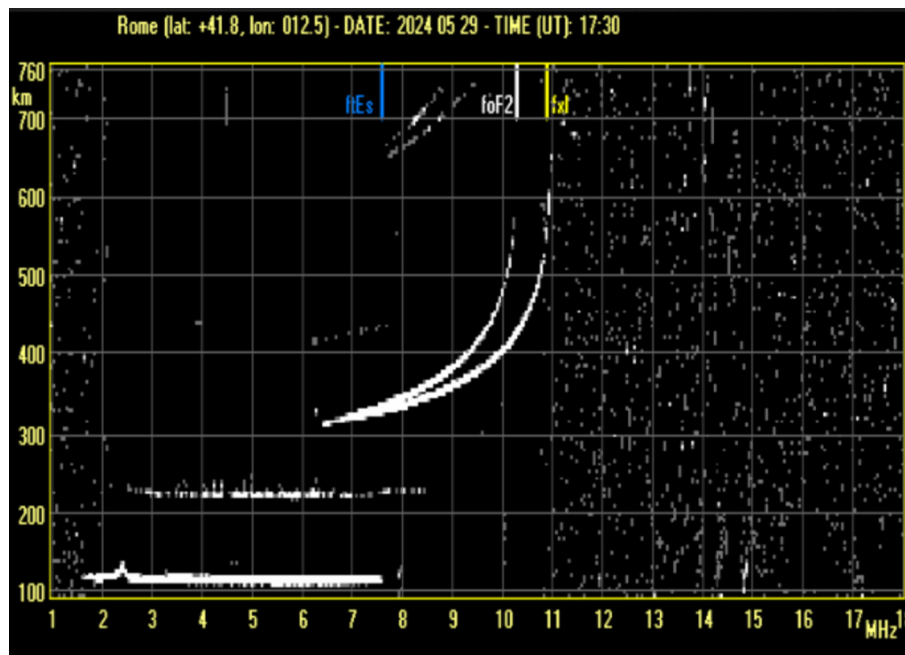


Fig. 16. AIS ionogram at 17:30 pm on 5/29/2024.

Since the reflected power from the ionosphere at all frequencies is less than 0 dBm, with a limiter at -4 dBm (Mini-Circuits PLS-01) in the front end circuit, the system can be safely used at maximum gain without entering a nonlinear zone. Furthermore, this parameter can be adjusted at any time to ensure good results and accurate scaling.

A series of tests was subsequently carried out using antennas of the ionospheric observatory of Rome (web reference: [Ionospheric Station of Rome](#)), to enable various radio soundings at different times. At the observatory, two ionosondes currently conduct radio soundings at intervals of 15 min each, without overlapping, each one in an assigned time slot out of the four available. The AISP Ionosonde tests began at the 22nd minute of each hour. The Ionogram from 17:24 pm (Fig. 15) on May 29, 2024, is compared with the closest ionogram from the AIS Ionosonde at the Rome observatory (reported respectively in Figs. 15 and 16).

The lower frequency component is visible in both ionograms. When compared to noise, the AISP performs better than the AIS, and the scattered signals are more clearly visible in the AISP. Even though an ionogram is considered to be static for a few seconds, the difference in the plotting time between the two is more than a minute, so slight differences in the plots can be observed.

8. Conclusions and developments

This work describes the project for constructing an ionosonde for vertical and oblique soundings, applicable for local and remote mapping, and analysis of the electronic content of the ionosphere.

The Ionosonde uses programmable Software Defined Radio (SDR) devices that allow various transmission and reception functions for radio frequency signals to be implemented via software. The use of these devices makes the system fully programmable from the command line. It can transmit and receive pulses encoded in various phases, ranging from simple pulsed signals to signals encoded with simple or complementary numerical codes (such as Barker and Golay codes). This programmability enables the ionosonde to perform radio sounding experiments applying different transmission techniques, and then compare the results in a very short time.

Looking at the Table A1, it is possible to note that most of the parameters are similar to those that characterize other similar systems. The most important innovations of AISP with respect to the AIS ionosonde are: a) the dimensions of the hardware, drastically reduced; b) the programmability of almost all the working parameters, that allows an easy change of the way of working, adapting them to changing operative conditions; 3) the introduction of a new algorithm for the identification of the peaks corresponding to a reflecting layer.

A possible future development of the system could involve the implementation of a GPSDO (GPS with Disciplined Oscillator) system, allowing synchronization with other ionosondes at remote sites in order to perform oblique surveys. This mode is easily achievable since the SDR devices are already equipped with these oscillators.

The ionograms produced by the AISP include a color intensity scale, with each color indicating a received signal intensity. However, even if the SDR system is only calibrated for the antenna and cable attenuation, the receiver

is still fully calibrated. Additionally, using antenna gain and RF power amplifier gain, the transmitter power could be calibrated and stored to compare the absorption properties of the transmission medium. All of these values are generated using software and can be used to define an effective ionogram.

The use of 2 SDR devices with 2 orthogonal receiver antennas will enable separation of the two polarizations of ordinary and extraordinary received rays in ionograms. These features will promote study of the details of ionospheric dynamics, extending the applications for this ionosonde.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. The characteristics of the AISP Ionosonde are summarized in Table A1.

Table A1
Characteristics of the AISP Ionosonde.

Parameter	Symbol	Value	Notes
Probing frequency range	f	1–18 MHz	or 2–15 MHz
Interval between probed frequencies	NF	50 kHz	
Number of tracks	f/NF	341	
Pulse repetition frequency	PRF	60 Hz	it can vary between 30 and 60 Hz
Pulse repetition interval	$1/PRF$	16.7 ms	transmitting two pulses per code the interval is halved to 8.3 ms
Pulse length (uncompressed)		480 μ s	“blind” distance $\approx$ 72 km
Integration factor	M	25	it can vary between 20 and 40
Persistence time at the same frequency		0.5 s	
Survey duration	T_{sound}	141 s	About 2.5 min. $T_{sound} = (\Delta f \cdot M)/(NF \cdot PRF)$
Code length for compression		16 bits	complementary code (2×16 bit)
Sub pulse length	τ	30 μ s	resolution = 4.5 km
Sampling rate	fs	200 kHz	
Number of samples per trace	Our	1000	
Echo reception time	Ns/fs	5.0 ms	or 2–15 MHz

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